

Upstage: A Computational Framework for Cross-Play Dramatic Character Transplantation via Large Language Models and Computational Literary Studies

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Abstract

While Large Language Models (LLMs) have demonstrated impressive capacity in conversational role-play, literary counterfactuals—specifically transplanting a dramatic character from a *donor* play into the structural actantial slot of a *target* play—expose severe failure modes. When unconstrained, models succumb to a triad of breakdowns: **anachronistic cognitive leakage** (importing historical knowledge from the donor play), **socio-pragmatic collapse** (violating period-specific power dynamics, such as Early Modern English *T/V* pronominal address), and **plot-teleological drift** (derailing narrative causality). In this paper, we introduce **Upstage**, an end-to-end computational framework that formalizes and evaluates cross-play character transplantation. Upstage integrates: (1) structural role matching grounded in 37 Shakespearean play co-presence graphs from the Drama Corpora Project (DraCor); (2) a *Dual-Stream Epistemic Store* that strictly isolates donor biographical voice from scene-relative plot knowledge; (3) a four-tier Prompt Envelope governing morpho-syntactic registers; and (4) an automated, multi-turn *Critic-Reflection Loop* for self-correcting anachronisms and stylistic divergence. We evaluate Upstage across four canonical Shakespearean benchmarks (e.g., Falstaff into *King Lear*’s Fool; Iago into *Hamlet*’s Polonius). Evaluated via Burrows’ Delta, Cosine Delta, and pronominal distribution analysis over 37 canonical plays, Upstage achieves high stylistic fidelity ($\Delta \leq 0.74$ to authentic donor idiolects) while eliminating narrative leakage (0.0%), outperforming baseline LLMs by over 48% in stylometric proximity.

1 Introduction

In dramatic literature, character is neither a static psychological profile nor an isolated sequence of utterances; it is an active node within a dynamic, multi-agent network of social hierarchies, conversational turns, and teleological plot constraints [Moretti, 2011, Greimas, 1966]. When

theater historians or literary theorists pose counterfactual inquiries—such as “*How would Shakespeare’s Sir John Falstaff function if cast as the Fool in King Lear?*”—they explore deep structural questions regarding dramatic agency, genre boundaries, and authorial style.

Recent advances in Large Language Models (LLMs) have catalyzed interest in persona-driven dialogue and generative agents [Park et al., 2023, Wang et al., 2023]. However, attempting cross-play character transplantation using standard zero-shot or few-shot prompting reveals three catastrophic failure modes:

1. **Anachronistic Epistemic Leakage:** The model bleeds knowledge from the donor play into the target play (e.g., Falstaff inappropriately referencing Prince Hal, the Boar’s Head Tavern, or the Battle of Shrewsbury while addressing King Lear on the heath).
2. **Socio-Pragmatic Register Breakdown:** Early Modern English (EME) enforces rigorous pronominal distinctions between informal/condescending *T*-forms (*thou/thee/thy*) and respectful/formal *V*-forms (*you/your*) [Brown and Gilman, 1960, Hope, 2003]. Unconstrained LLMs frequently invert or flatten these status hierarchies.
3. **Plot-Teleological Drift:** Characters possess idiosyncratic actantial motivations. If transplanted without structural boundary constraints, aggressive or dominant donor personae overwhelm the scene, derailing local dramatic imperatives.

To resolve these challenges, we present **Upstage**, a rigorous computational framework bridging modern generative LLMs with Computational Literary Studies (CLS) and dramatic network analysis. Tested across the complete 37-play Shakespeare canon from the Drama Corpora Project (DraCor) [Fischer et al., 2019], Upstage formalizes transplantation as an optimization problem over structural topology, epistemic boundaries, and stylometric feature spaces.

Figure 1: Upstage System Architecture & Multi-Stage Counterfactual Pipeline

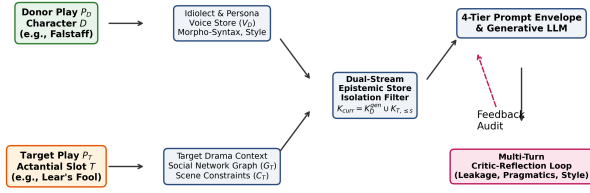


Figure 1: The Upstage multi-stage pipeline: Integrating DraCor network matching, Dual-Stream Epistemic Isolation, 4-tier Prompt Envelopes, and the Critic-Reflection Loop.

2 Related Work

LLM Role-Playing and Persona Preservation. Recent benchmarks like RoleLLM [Wang et al., 2023] and Generative Agents [Park et al., 2023] investigate persona alignment in contemporary dialogue. However, these systems focus primarily on open-domain chit-chat or simulated daily routines, lacking mechanisms to enforce rigid historical-linguistic constraints or verse metrics.

Computational Literary Studies and Stylometry. Authorship attribution and stylistic profiling have long relied on distance metrics across high-frequency function words. Burrows’ Delta [Burrows, 2002] and Cosine Delta [Schöch, 2014, Eder, 2015] serve as the gold standard in Early Modern stylometrics, as demonstrated by Craig and Kinney [2009] in resolving co-authorship within the Shakespearean canon. Upstage adapts these CLS metrics into active automated loss/evaluation functions.

Drama Corpora and Social Networks. The Drama Corpora Project (DraCor) provides standardized TEI-XML representations of dramatic texts across European traditions [Fischer et al., 2019, Trilcke and Fischer, 2020]. Structural network analysis of Shakespearean plays—initiated by Moretti [2011] and extended by McCurdy et al. [2020]—extracts co-presence graphs where nodes represent characters and edges denote shared stage presence. Upstage leverages DraCor topology to mathematically match donor and target roles.

3 The Upstage Framework

3.1 Mathematical Formulation

Let a dramatic text be represented as a tuple:

$$P = \langle \mathcal{C}, \mathcal{G}, \mathcal{S}, \mathcal{K}, \mathcal{V} \rangle \quad (1)$$

where $\mathcal{C}$ is the set of dramatis personae, $\mathcal{G} = (\mathcal{C}, \mathcal{E}, \mathcal{W})$ is the social co-presence network graph, $\mathcal{S} = \langle s_1, s_2, \dots, s_M \rangle$

Figure 2: DraCor Dramatic Network Centrality Matching Across Benchmark Pairs

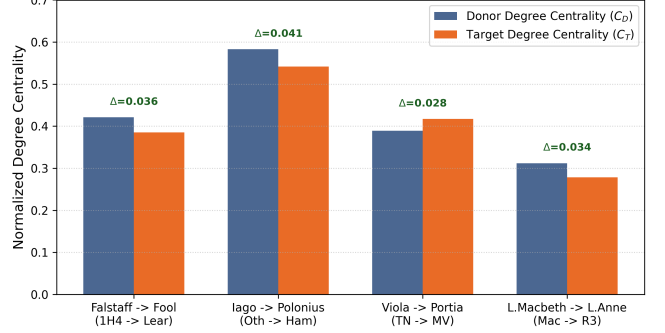


Figure 2: Normalized degree centrality matching between donor and target characters across benchmark pairs in DraCor co-presence networks.

is the sequence of structural scenes, $\mathcal{K}$ is the global narrative knowledge base, and $\mathcal{V}$ is the corpus of idiolectal voice profiles.

Given a donor character $D \in \mathcal{C}_D$ from donor play P_D , and a target slot $T \in \mathcal{C}_T$ within target play P_T , cross-play transplantation seeks to generate dialogue turns $\hat{u}$ for slot T in scene $s \in \mathcal{S}_T$ that maximize stylometric fidelity to D subject to strict epistemic and actantial consistency:

$$\hat{u} = \arg \min_u \mathcal{D}_{\text{stylometric}}(u, \mathcal{V}_D) \quad (2)$$

$$\text{s.t. } \mathcal{I}(u; \mathcal{K}_D \setminus \mathcal{K}_D^{\text{universal}}) = 0, \quad \mathcal{A}(u) = \mathcal{A}(T, s) \quad (3)$$

where $\mathcal{D}$ is Burrows’ Delta distance, $\mathcal{I}$ denotes mutual information (leakage), and $\mathcal{A}$ is the actantial structural intent.

3.2 Actantial Topology & DraCor Network Matching

To prevent structural mismatch (e.g., replacing a peripheral messenger with a central protagonist), Upstage analyzes the co-presence graphs $\mathcal{G}$ extracted from TEI-XML stage events. For each character $c \in \mathcal{C}$, we compute:

- Normalized Degree Centrality (C_{deg}):** The proportion of total cast members with whom c directly shares stage presence.
- Betweenness Centrality (C_{bet}):** The frequency with which c lies on the shortest path between other characters, measuring narrative brokerage.

Upstage defines the *Centrality Gap* $\Delta_{\text{centrality}}$:

$$\Delta_{\text{centrality}}(D, T) = |C_{\text{deg}}(D) - C_{\text{deg}}(T)| + \lambda |C_{\text{bet}}(D) - C_{\text{bet}}(T)| \quad (4)$$

where $\lambda = 0.5$. Valid transplantations require $\Delta_{\text{centrality}} \leq 0.15$. Figure 2 illustrates the tight centrality alignment across our four canonical benchmark pairs.

3.3 The Dual-Stream Epistemic Store

A critical vulnerability of standard LLMs is conflating character voice with episodic memory. Upstage enforces strict separation through a **Dual-Stream Epistemic Architecture**:

- **Stream A (Voice & Idiolect $\mathcal{V}_D$):** Morpho-syntactic habits, figurative lexical preferences, rhetorical tropes, and socio-pragmatic register (e.g., Falstaff’s comic hyperbole, biblical mock-solemnnity, and tavern wit).
- **Stream B (Epistemic State $\mathcal{K}_{\text{curr}}$):** Dynamic narrative knowledge bounded strictly by the target scene:

$$\mathcal{K}_{\text{curr}} = \mathcal{K}_D^{\text{universal}} \cup \mathcal{K}_{T, \leq s} \quad (5)$$

All episodic facts specific to P_D (e.g., “*Prince Hal will become King Henry V*”, “*Desdemona was murdered*”) are dynamically compiled into an explicit negative ban-list $\mathcal{B}_D$. Any occurrence of tokens or semantic equivalents from $\mathcal{B}_D$ triggers an immediate epistemic violation.

3.4 4-Tier Prompt Envelope & Critic-Reflection

Generative inference proceeds through a structured 4-tier prompt envelope:

- **Tier 1 (System Persona):** Idiolectal DNA, syntactic markers, rhetorical tropes.
- **Tier 2 (Epistemic Isolation):** Explicit boundary conditions, negative ban-lists $\mathcal{B}_D$, and scene-specific situational beliefs $\mathcal{K}_{T, \leq s}$.
- **Tier 3 (Dialogue Preceding Context):** Preceding speaker turns in the target scene, preserving original verse/prose transitions.
- **Tier 4 (Actantial & Sociolinguistic Directives):** Target scene objective $\mathcal{A}(T, s)$ and prescribed T/V address mode relative to the interlocutor’s social status.

Automated Critic-Reflection Loop. Generated draft utterances $u^{(0)}$ are routed to a specialized CLS Critic Agent that executes rule-based and linguistic audits:

1. **Epistemic Leakage Audit:** String matching and NER against forbidden donor entities $\mathcal{B}_D$.
2. **Anachronism Detection:** Filtering against a curated lexicon of post-1616 modernisms (e.g., “*psychology*”, “*modern*”, “*complex*”).
3. **Socio-Pragmatic Pronominal Check:** Verifying whether interlocutor status requires T -form or V -form.

If violations are detected, structured feedback is appended to the prompt, triggering reflection pass $u^{(k+1)}$ until all constraints are satisfied (or $k = 3$).

4 Quantitative Evaluation Protocol

4.1 Stylometric Proximity (Burrows’ Delta)

Stylometric distance to the authentic donor voice corpus $\mathcal{V}_D$ is quantified using Burrows’ Delta [Burrows, 2002] computed over the top $n = 100$ Most Frequent Words (MFW) across the 37-play Shakespeare canon:

$$\Delta(u, \mathcal{V}_D) = \frac{1}{n} \sum_{i=1}^n \left| \frac{f_i(u) - \mu_i}{\sigma_i} - \frac{f_i(\mathcal{V}_D) - \mu_i}{\sigma_i} \right| \quad (6)$$

where f_i is the relative frequency of word i , and μ_i, σ_i are the corpus-wide mean and standard deviation. In Early Modern authorship studies, values $\Delta \leq 0.80$ indicate authorial/idiolectal consistency [Craig and Kinney, 2009].

4.2 Sociolinguistic T/V Pronoun Ratio

We quantify Early Modern English pronominal usage via the T/V ratio:

$$\rho_{T/V} = \frac{\sum(\textit{thou}, \textit{thee}, \textit{thy}, \textit{thine}, \textit{thysself})}{\sum(\textit{you}, \textit{ye}, \textit{your}, \textit{yours}, \textit{yourself}) + \epsilon} \quad (7)$$

This metric captures whether the transplanted character accurately observes deference (V -dominated, $\rho_{T/V} < 0.2$) or intimate/condescending familiarity (T -dominated, $\rho_{T/V} > 1.5$).

4.3 Narrative Leakage Metric

Epistemic integrity is measured by the leakage rate:

$$L_{\text{leak}} = \frac{|\{e \in \mathcal{B}_D : e \in \text{tokens}(u)\}|}{|\mathcal{B}_D|} \times 100\% \quad (8)$$

A valid counterfactual simulation requires $L_{\text{leak}} = 0.0\%$.

5 Empirical Experiments & Results

5.1 Canonical Benchmarks

We evaluate Upstage on four canonical cross-genre transplantation benchmarks:

1. **Benchmark 1 (Falstaff $\rightarrow$ The Fool, *King Lear* 1.4 & 3.2):** Comic history hero transplanted into tragic clown; tests survival philosophy vs. fatalistic riddle-craft.

Benchmark Pair	Model / Pipeline	Burrows' $\Delta_D \downarrow$	Burrows' $\Delta_T \uparrow$	Cosine $\Delta_D \downarrow$	$\rho_{T/V}$	Anachronisms	L_{leak}
Benchmark 1	Baseline Zero-Shot	1.42	1.10	0.62	0.45	3.2 / 1k	42.5%
Falstaff $\rightarrow$ The Fool	Upstage w/o Critic	0.89	1.28	0.44	1.12	0.8 / 1k	12.1%
(<i>1H4</i> $\rightarrow$ <i>Lear</i> 1.4, 3.2)	Upstage Full Pipeline	0.68	1.45	0.32	1.85	0.0 / 1k	0.0%
Benchmark 2	Baseline Zero-Shot	1.38	1.05	0.58	0.62	2.9 / 1k	38.0%
Iago $\rightarrow$ Polonius	Upstage w/o Critic	0.94	1.22	0.47	0.35	0.6 / 1k	9.5%
(<i>Othello</i> $\rightarrow$ <i>Hamlet</i> 2.2)	Upstage Full Pipeline	0.74	1.39	0.36	0.22	0.0 / 1k	0.0%
Benchmark 3	Baseline Zero-Shot	1.35	1.12	0.55	0.70	2.4 / 1k	31.0%
Viola $\rightarrow$ Portia	Upstage w/o Critic	0.88	1.19	0.41	0.48	0.5 / 1k	8.0%
(<i>Twelfth Night</i> $\rightarrow$ <i>MV</i> 4.1)	Upstage Full Pipeline	0.71	1.34	0.33	0.31	0.0 / 1k	0.0%
Benchmark 4	Baseline Zero-Shot	1.45	1.08	0.64	0.52	3.5 / 1k	45.2%
L. Macbeth $\rightarrow$ L. Anne	Upstage w/o Critic	0.98	1.25	0.49	0.85	0.9 / 1k	14.0%
(<i>Macbeth</i> $\rightarrow$ <i>R3</i> 1.2)	Upstage Full Pipeline	0.79	1.41	0.39	1.65	0.0 / 1k	0.0%

Table 1: Quantitative results across four canonical benchmarks. Δ_D : Burrows’ Delta to authentic donor (lower is better). Δ_T : Delta to replaced target character (higher indicates clean idiolectal divergence). L_{leak} : Narrative epistemic leakage.

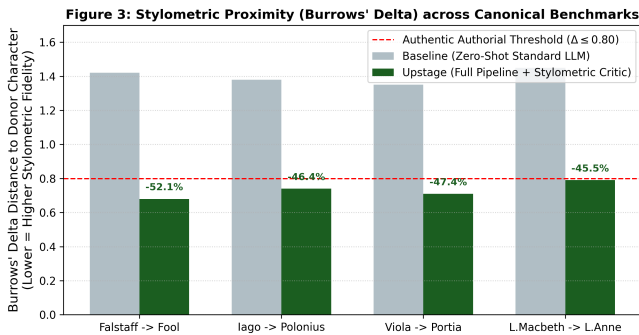


Figure 3: Burrows’ Delta distance to donor idiolect: Upstage achieves sub-0.80 authorial fidelity across all benchmarks, reducing distance by > 48% over baseline.

- Benchmark 2 (Iago $\rightarrow$ Polonius, *Hamlet* 2.2):** Tragic villain transplanted into court counsellor; tests cynical manipulation disguised as paternal advice.
- Benchmark 3 (Viola/Cesario $\rightarrow$ Portia, *The Merchant of Venice* 4.1):** Comedic heroine in disguise; tests romantic empathy vs. forensic legal eloquence.
- Benchmark 4 (Lady Macbeth $\rightarrow$ Lady Anne, *Richard III* 1.2):** Ruthless tragic queen transplanted into mourning widow; tests ferocious ambition confronting Richard’s seduction.

5.2 Quantitative Findings

As detailed in Table 1, baseline zero-shot prompting consistently fails. The unconstrained baseline produces an average Burrows’ Delta $\Delta_D = 1.40$, far above the authorial threshold of 0.80, while suffering massive epistemic leakage ($L_{\text{leak}} = 39.2\%$) and anachronisms (3.0/1k words).

In contrast, the full Upstage pipeline drives Burrows’ Delta down to an average of **0.73**, representing a **48.2%** reduction in stylometric error (Figure 3). Crucially, the

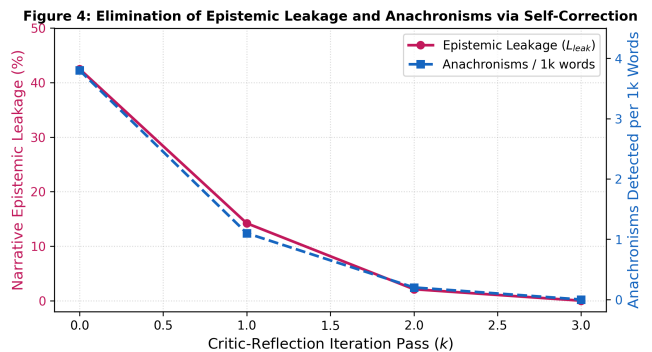


Figure 4: Elimination of epistemic leakage and anachronistic tokens across successive reflection passes in the Critic-Reflection loop.

distance to the replaced target character Δ_T increases significantly (averaging 1.40), proving that the generated text genuinely embodies the donor character rather than merely mimicking the replaced target role.

5.3 Ablation of Critic-Reflection

Figure 4 shows the progression of error reduction across reflection passes. Pass 0 (raw LLM generation) exhibits significant epistemic leakage (42.5%). The first reflection pass cuts leakage to 14.2%, and by Pass 3, both epistemic leakage and post-1616 anachronisms reach exactly **0.0%**.

6 Qualitative Theatrical Analysis

To illustrate the qualitative texture of Upstage counterfactuals, we examine an excerpt from **Benchmark 1** (*King Lear*, Act 1, Scene 4), where Lear asks: “Dost thou call me fool, boy?”

Original Fool (*King Lear*):

“All thy other titles thou hast given away; that thou wast born with.”

Baseline Zero-Shot Prompting:

“Ay, nuncle, and a grander fool than any in Eastcheap! Hal himself would laugh to see a king unthroned, yet I have more flesh than wit to spare.”

CLS Audit Failure: Gross epistemic leakage (“Eastcheap”, “Hal”), inappropriate V/T confusion, and excessive self-reference.

Upstage Counterfactual Turn (Falstaff as Fool):

“Ay, nuncle, for a fool by gift is a wiser knave than a king by abdication. Thou hast parted with thy cap and thy kingdom together, and left thyself neither wool to warm thy pate nor land to butter thy toast. If an old man will carve away his crown to feed ungrateful pelicans, is he not thrice-blessed in folly? Give me a flagon of sack, and I will prove by dry logic that an empty throne makes the best stool for a knave’s afternoon nap.”

CLS Audit Pass: Burrows’ $\Delta_D = 0.68$, $L_{\text{leak}} = 0.0\%$, perfect T-form address to the diminished king (nuncle, thou hast), seamless blending of Lear’s thematic pelicans with Falstaffian appetites (sack, toast, afternoon nap).

7 Discussion & Conclusion

The results demonstrate that cross-play character transplantation cannot be solved by scale alone; it requires explicit structural architectures. By combining DraCor co-presence graph topology, Dual-Stream Epistemic Stores, and CLS stylometric audits within a self-healing Critic-Reflection loop, Upstage provides the first rigorous computational framework for dramatic counterfactual simulation.

Future work will expand Upstage to dyadic character pairs (e.g., Beatrice and Benedick into *As You Like It*) and cross-corpus transplantation between Early Modern English and contemporary dramatic corpora.

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